IM101
Finals Task #3
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BSIT 3B

1. SQL statements for (Task 1-5):

```
DROP TABLE IF EXISTS employees;
CREATE TABLE employees (
  emp_id INT AUTO_INCREMENT PRIMARY KEY,
  emp_no VARCHAR(10) NOT NULL UNIQUE,
  first_name VARCHAR(60) NOT NULL,
  last_name VARCHAR(60) NOT NULL,
  email VARCHAR(100) UNIQUE,
  phone VARCHAR(30),
  hire_date DATE NOT NULL,
  job_title VARCHAR(80),
  salary DECIMAL(10,2) DEFAULT 0.00,
  manager_id INT DEFAULT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  INDEX idx_lastname_firstname (last_name, first_name)
) ENGINE=InnoDB;

ALTER TABLE employees
  ADD CONSTRAINT fk_employees_manager
  FOREIGN KEY (manager_id) REFERENCES employees(emp_id)
  ON DELETE SET NULL
  ON UPDATE CASCADE;

CREATE TABLE departments (
  dept_id INT AUTO_INCREMENT PRIMARY KEY,
  dept_code VARCHAR(10) NOT NULL UNIQUE,
  dept_name VARCHAR(100) NOT NULL,
  location VARCHAR(100),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB;

CREATE TABLE employee_dept (
  emp_dept_id INT AUTO_INCREMENT PRIMARY KEY,
  emp_id INT NOT NULL,
  dept_id INT NOT NULL,
  start_date DATE NOT NULL,
  end_date DATE DEFAULT NULL,
  role VARCHAR(80),
  CONSTRAINT fk_ed_emp FOREIGN KEY (emp_id) REFERENCES employees(emp_id) ON DELETE CASCADE,
  CONSTRAINT fk_ed_dept FOREIGN KEY (dept_id) REFERENCES departments(dept_id) ON DELETE CASCADE,
  UNIQUE KEY uq_emp_dept_period (emp_id, dept_id, start_date)
) ENGINE=InnoDB;

CREATE TABLE projects (
  project_id INT AUTO_INCREMENT PRIMARY KEY,
  project_code VARCHAR(20) NOT NULL UNIQUE,
  project_name VARCHAR(150),
  start_date DATE,
  end_date DATE,
  budget DECIMAL(12,2) DEFAULT 0.00
) ENGINE=InnoDB;
```

```
CREATE TABLE employee_projects (
  emp_proj_id INT AUTO_INCREMENT PRIMARY KEY,
  emp_id INT NOT NULL,
  project_id INT NOT NULL,
  assigned_date DATE NOT NULL,
  role_on_project VARCHAR(100),
  hours_allocated DECIMAL(7,2) DEFAULT 0.00,
  CONSTRAINT fk_ep_emp FOREIGN KEY (emp_id) REFERENCES employees(emp_id) ON DELETE CASCADE,
  CONSTRAINT fk_ep_proj FOREIGN KEY (project_id) REFERENCES projects(project_id) ON DELETE CASCADE,
  UNIQUE KEY uq_emp_proj (emp_id, project_id)
) ENGINE=InnoDB;

CREATE TABLE managers (
  manager_id INT PRIMARY KEY,
  manager_level INT NOT NULL DEFAULT 1,
  title VARCHAR(100),
  since DATE,
  CONSTRAINT fk_managers_emp FOREIGN KEY (manager_id) REFERENCES employees(emp_id) ON DELETE
CASCADE
) ENGINE=InnoDB;
```

2. Table Structure (Employees, departments, employee_dept, employee_projects and managers) – Following data are dummy datas.

A. Department Table

	dept_id	dept_code	dept_name	location	created_at
▶	1	HR	Human Resources	Manila	2025-10-29 11:17:15
	2	FIN	Finance	Manila	2025-10-29 11:17:15
	3	ENG	Engineering	Cebu	2025-10-29 11:17:15
	4	MKT	Marketing	Manila	2025-10-29 11:17:15
	5	OPS	Operations	Davao	2025-10-29 11:17:15
*	NULL	NULL	NULL	NULL	NULL

B. Employee Department Table

	emp_dept_id	emp_id	dept_id	start_date	end_date	role
▶	1	1	1	2016-05-02	NULL	Executive
	2	1	2	2016-05-02	NULL	Executive
	3	2	2	2017-03-15	NULL	Finance Lead
	4	3	3	2018-07-22	NULL	Tech Lead
	5	4	3	2019-02-10	NULL	Engineering Manager
	6	5	3	2020-01-05	NULL	Senior Engineer
	7	6	3	2021-06-30	NULL	Engineer
	8	7	1	2019-08-01	NULL	HR Manager
	9	8	1	2022-04-20	NULL	HR Specialist
	10	9	4	2023-09-01	NULL	Marketing Lead
	11	10	5	2024-02-11	NULL	Ops Manager
*	NULL	NULL	NULL	NULL	NULL	NULL

C. Employee Projects Table

emp_proj_id	emp_id	project_id	assigned_date	role_on_project	hours_allocated
1	3	1	2023-01-10	Architect	120.00
2	4	1	2023-02-01	Lead Engineer	800.00
3	5	1	2023-03-01	Senior Engineer	1000.00
4	6	1	2023-05-01	Engineer	600.00
5	2	2	2024-03-05	Finance Analyst	200.00
6	8	3	2024-05-15	HR Analyst	150.00
7	7	3	2024-05-01	Project Sponsor	100.00
NULL	NULL	NULL	NULL	NULL	NULL

D. Employees Table

emp_id	emp_no	first_name	last_name	email	phone	hire_date	job_title	salary	manager_id	created_at	updated_
1	E0001	Aileen	Dizon	aileen.dizon@example.com	09171234567	2016-05-02	CEO	200000.00	NULL	2025-10-29 11:17:15	2025-10-2
2	E0002	Marco	Reyes	marco.reyes@example.com	09171234568	2017-03-15	CFO	150000.00	1	2025-10-29 11:17:15	2025-10-2
3	E0003	Liza	Garcia	liza.garcia@example.com	09171234569	2018-07-22	CTO	150000.00	1	2025-10-29 11:17:15	2025-10-2
4	E0004	Ramon	Santos	ramon.santos@example.com	09171234570	2019-02-10	Head of Engineering	120000.00	3	2025-10-29 11:17:15	2025-10-2
5	E0005	Joy	Torres	joy.torres@example.com	09171234571	2020-01-05	Senior Engineer	80000.00	4	2025-10-29 11:17:15	2025-10-2
6	E0006	John	Lacson	john.lacson@example.com	09171234572	2021-06-30	Engineer	60000.00	4	2025-10-29 11:17:15	2025-10-2
7	E0007	Rosa	Mendoza	rosa.mendoza@example.com	09171234573	2019-08-01	Head of HR	110000.00	2	2025-10-29 11:17:15	2025-10-2
8	E0008	Peter	Cruz	peter.cruz@example.com	09171234574	2022-04-20	HR Specialist	45000.00	7	2025-10-29 11:17:15	2025-10-2
9	E0009	Anna	Lopez	anna.lopez@example.com	09171234575	2023-09-01	Marketing Lead	90000.00	1	2025-10-29 11:17:15	2025-10-2
10	E0010	Ben	Ibarra	ben.ibarra@example.com	09171234576	2024-02-11	Operations Manager	95000.00	1	2025-10-29 11:17:15	2025-10-2

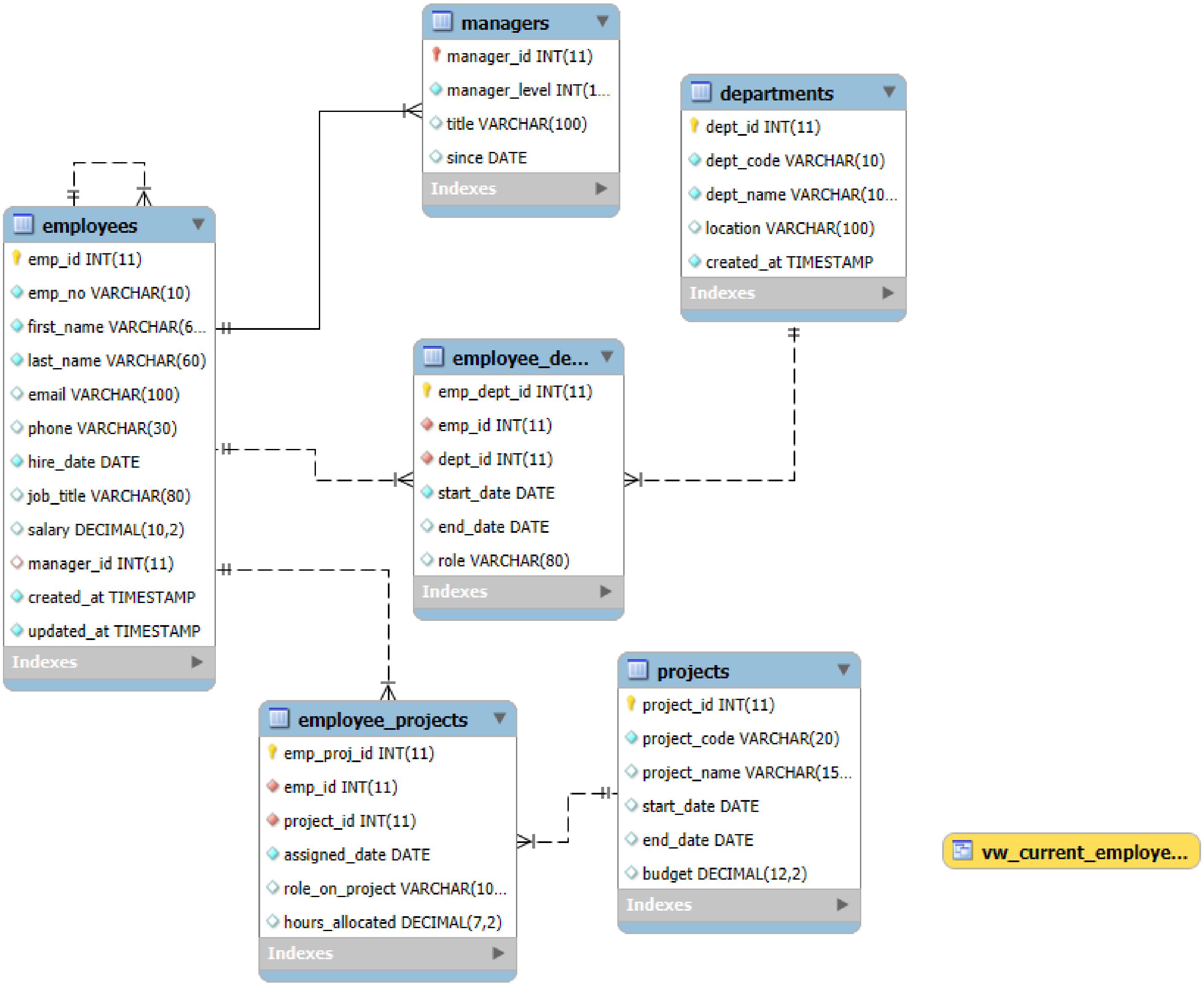
E. Managers Table

manager_id	manager_level	title	since
1	3	CEO	2016-05-02
2	2	CFO	2017-03-15
3	2	CTO	2018-07-22
4	1	Head of Engineering	2019-02-10
7	1	Head of HR	2019-08-01
9	1	Marketing Lead	2023-09-01
10	1	Operations Manager	2024-02-11
NULL	NULL	NULL	NULL

F. Projects Table

project_id	project_code	project_name	start_date	end_date	budget
1	P-ALPHA	Alpha Platform	2023-01-01	NULL	500000.00
2	P-BETA	Beta Release	2024-03-01	2024-12-31	200000.00
3	P-HR2024	HR Automation	2024-05-01	2025-05-01	80000.00
NULL	NULL	NULL	NULL	NULL	NULL

3. Relational schema from the EER model of Mysql Workbench (pdf or jpg file)



managers
manager_id INT(11)
manager_level INT(1...
title VARCHAR(100)
since DATE
Indexes
departments
dept_id INT(11)
dept_code VARCHAR(10)
dept_name VARCHAR(10...
location VARCHAR(100)
created_at TIMESTAMP
Indexes
employees
emp_id INT(11)
emp_no VARCHAR(10)
first_name VARCHAR(6...
last_name VARCHAR(60)
email VARCHAR(100)
phone VARCHAR(30)
hire_date DATE
job_title VARCHAR(80)
salary DECIMAL(10,2)
manager_id INT(11)
created_at TIMESTAMP
updated_at TIMESTAMP
Indexes
employee_de...
emp_dept_id INT(11)
emp_id INT(11)
dept_id INT(11)
start_date DATE
end_date DATE
role VARCHAR(80)
Indexes
projects
project_id INT(11)
project_code VARCHAR(20)
project_name VARCHAR(15...
start_date DATE
end_date DATE
budget DECIMAL(12,2)
Indexes
employee_projects
emp_proj_id INT(11)
emp_id INT(11)
project_id INT(11)
assigned_date DATE
role_on_project VARCHAR(10...
hours_allocated DECIMAL(7,2)
Indexes
vw_current_employe...